Krishnaa **Vadivel** | GPU Software Engineer | High-Performance Computing

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Summary

GPU-focused computational scientist and software engineer for performance-oriented systems and HPC roles. Builds Python and C++ code with CUDA, ROCm, MPI, and OpenMP, with experience in multi-GPU parallelization, collective communication, distributed debugging, and strong and weak scaling analysis on national-lab supercomputers.

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL
GPU / HPC: CUDA, ROCm, MPI, OpenMP, collective communication, multi-GPU parallelization, PETSc, SLEPc
Platforms: AMD MI250X, NVIDIA A100, Frontier, Perlmutter, Linux, CMake, Docker, Kubernetes
Debug / Perf: TotalView, Linaro DDT, distributed debugging, strong scaling, weak scaling, hardware-aware optimization
Machine Learning: PyTorch, PyTorch Geometric, automatic differentiation, equivariant graph neural networks, scikit-learn

Experience

Yale University, Electrical Engineering
PhD Research Assistant *2021–present*

- Developed distributed scientific software in Python and C++ for first-principles simulation workloads as part of an INCITE award using exascale resources, with MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier at OLCF and Perlmutter at LBNL.
- Parallelized matrix and vector workloads across multiple GPUs, using collective communication and distributed linear-algebra workflows to scale simulations across clusters.
- Ported and optimized scientific-computing workloads for AMD MI250X and NVIDIA A100 GPU systems, translating many-body and excited-state physics methods into scalable GPU implementations.
- Built PyTorch and PyTorch Geometric models, including equivariant graph neural networks, that delivered roughly 100x speedups for parts of the workflow by combining model training with automatic differentiation to obtain additional quantities without separate expensive calculations.
- Ran strong- and weak-scaling studies to evaluate distributed performance and used TotalView, CUDA-aware MPI workflows, and Linaro DDT to debug multi-process and multi-GPU applications across heterogeneous compute environments.
- Packaged scientific software in Docker containers for reproducible deployment across local and HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS research in 2024, 2025, and 2026, grounding GPU and HPC work in production research workflows.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built engineering software and supporting backend tooling for real-time remote tool testing, data collection, and sensor calibration workflows.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight
Photonics Technical Writer *2018–2019*

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Multi-GPU Exciton-Phonon Simulation: Developed distributed first-principles calculations with CUDA, ROCm, MPI, and OpenMP, including multi-GPU matrix and vector parallelization on leadership-class GPU clusters.

Scientific ML on GPU Systems: Developed PyTorch Geometric equivariant models that produced roughly 100x workflow speedups by pairing training with automatic differentiation to recover additional quantities efficiently.

CUDA Containers for HPC Software: Packaged scientific applications in CUDA-ready Docker environments for portable deployment across local and HPC systems.

Matrix-Multiply AI Accelerator: Built a Verilog-based matrix-multiply accelerator project for AI-oriented datapath design and hardware-aware compute fundamentals.

Education

Yale University
PhD, Electrical Engineering — *2021–present*

Yale University
MPhil, Electrical Engineering — *2023*

Yale University
MS, Electrical Engineering — *2023*

Cornell University
MEng, Electrical and Computer Engineering — *2017–2018*

Cornell University
BS, Electrical and Computer Engineering — *2013–2017*